DSA Test - Medium Level

Instructions

- Time: 75 minutes
- Total Questions: 5
- No external references allowed

Question 1 - Pattern (Medium)

Write a program to print a **diamond pattern** for `n = 5`:

```
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *
```

Input: An integer `n` (number of rows in the upper half)

Output: The diamond star pattern

Question 2 - Pattern (Medium)

Write a program to print **Pascal's Triangle** up to `n = 5` rows:

```
    1
   1 1
  1 2 1
 1 3 3 1
1 4 6 4 1
```

Input: An integer `n`

Output: Pascal's triangle with proper spacing

Question 3 - Hashing (Medium)

Given an array of integers and a target sum `k`, find the **length of the longest subarray** whose elements sum to `k`.

Example:

Input: `arr = [10, 5, 2, 7, 1, 9]`, `k = 15`

Output: `4` (subarray `[5, 2, 7, 1]`)

Constraints:

- 1 ≤ arr.length ≤ 10^5
- -10^5 ≤ arr[i] ≤ 10^5

Hint: Use prefix sum with a hash map.

Question 4 - Array (Medium)

Given an array of `n` integers, **rearrange** the array in alternating positive and negative numbers while maintaining relative order.

Example:

Input: `[1, 2, -3, -1, -2, 3]`

Output: `[1, -3, 2, -1, 3, -2]`

Constraints:

- The array has equal number of positive and negative integers.
- 1 ≤ arr.length ≤ 10^5

Question 5 - Matrix (Medium)

Given an `n x n` matrix, **rotate it 90 degrees clockwise** in-place.

Example:

Input:

```
1   2   3
4   5   6
7   8   9
```

Output:

```
7   4   1
8   5   2
9   6   3
```

Constraints:

- $1 \le n \le 500$
- Do it in O(1) extra space.